

How CC Patio models must be built

SketchUp modelling standard for frames · DRAFT — NOT ADOPTED

A script reads the SketchUp model and produces the part numbers, the bill of materials, the cut list and the material cost. **It measures geometry. It does not read your component names, and it cannot ask you what you meant.**

That makes the model the source data rather than a picture. A rail drawn 12" short does not look wrong on screen — it produces a cut list that is 12" short, and the steel is cut before anyone notices.

**THE MODEL IS THE SPEC.
IF IT IS NOT IN THE MODEL, IT DOES NOT EXIST.**

What comes out of a conforming file

Every part gets a number computed from the steel itself — profile, length, end cuts. Nobody assigns it, nobody maintains a list, and the same physical piece measured in any model produces the same number.

```
CUT- SQ2x16 - 034.0 - 45 45 C
|          |          | | +-- both ends angled: C converging, S same direction
|          |          +---+----- end cut angles, degrees
|          +----- length in inches, LONG POINT
+----- stock profile: 2x2 square tube, 16 ga
```

Then the bill of materials, with real quantities, straight from the model:

```
BRAVADA CLUB CHAIR 34 x 34
├─ CUT-SQ2x16-034.0-4545C    x5   seat frame rail x4 + back top rail x1
├─ CUT-SQ2x16-008.0-9090    x4   leg
├─ CUT-FB0.1875x2-002.0-9090 x4   leg cap
├─ CUT-FB0.125x1-030.0-9090 x5   seat cross brace
├─ CUT-SQ2x16-032.0-4590    x2   arm top rail
├─ CUT-SQ2x16-014.0-9045    x2   arm front post
├─ CUT-SQ2x16-019.0-9045    x2   back post
├─ CUT-SQ2x16-030.0-9090    x1   back mid rail
├─ CUT-RT1.5x0.75x16-030.0-9090 x1 seat centre rail

SQ2x16 total  30.17 ft
```

Note the first line. The seat frame rail and the back top rail are the same part — same steel, same length, same cut. In the old model they were named SEAT FRAME 34" and BACK TOP 30", filed pages apart, and nobody had ever

noticed. Finding that is the point of the exercise: one saw setup instead of two, and the highest-count part in the chair.

The rules

STRUCTURE

1 Every physical part is its own component

One component equals one thing that comes off the saw. Not loose geometry, and never two physical parts merged into one component.

2 Identical parts must be the same component, copied

Draw the leg once, then copy that component. **Do not redraw it, and do not explode and rebuild it.** Four legs should be four copies of one component, not four separate components that happen to look alike.

3 Sub-assemblies get their own component

Arm, back, seat frame — anything welded up as a unit before final assembly. This is what gives the bill of materials structure instead of a flat list, and it is what lets a shared weldment (the same arm on a chair, a loveseat and a sofa) be recognised once and batch-built.

4 Wrap the whole product in one top-level component

Named for the product, with everything else inside it.

```
BRAVADA COFFEE TABLE 42 x 42
```

Without this the parts are readable but the product is not, and someone has to assert by hand what was drawn.

GEOMETRY — THE TWO THAT ARE INVISIBLE ON SCREEN

5 Draw each part square to the axes inside its own component, then rotate it into place

The script measures each component in its own local space. **A part drawn at an angle inside its own component measures wrong** — the length comes out as the diagonal of a box instead of the length of the part.

```
RIGHT  draw the rail flat along an axis, make it a component,  
        THEN rotate the copy into position
```

```
WRONG  draw the rail already tilted in space and make that a component
```

6 Draw parts at their true cut size — and length means the long point

On a mitred part the stated length is the **longest edge** of the finished piece. Always. A 45° mitre on 2" tube moves the length by 2" per angled end, so a part drawn short-point produces a cut list 2–4" short. That is scrap, not a rounding error.

7 Do not name anything

Component names are ignored entirely. Component#1 is fine. The part number is computed from the steel, so a name adds nothing — and a wrong name is worse than no name. In the existing club chair model, BACK TOP was named 30" and measures 34"; ARM VERTICAL was named 90/90 and measures 90/45.

There is exactly one exception: gauge. Steel is assumed 16 ga. If a part is not, put the gauge in that component's name and change nothing else:

11 ga	11 GA	11g	.120
-------	-------	-----	------

8 Model every part, including the small ones

Caps, gussets, tabs, brackets. If it is cut or welded, it is a part and it belongs in the model. A cap that is not modelled is a cap that is never bought — the club chair has four and they were missing from the bill of materials entirely.

9 Purge unused components before sending the file

Window > Model Info > Statistics > Purge Unused. Left-over definitions that are not placed anywhere inflate the part list and produce false duplicate warnings.

What breaks it

Every one of these was found in a real CC Patio file. **None are visible in the Outliner, and most are not visible on screen either.**

FOUND IN THE FILE	WHAT IT PRODUCES
A 108" sofa frame with two legs, both at the front	No back support. Cut list missing four legs.
No front or back seat rail — two end rails joined by nothing	The frame cannot be built from its own bill of materials.
Five flat bars 92" long in a 104" clear span	Twelve inches short of the rail they must weld to. Scrap.
Five slats landing on nothing at either end	Nothing to weld to.
An assembly placed nowhere , left over from earlier work	Inflated part count and false duplicate warnings.
The same weldment drawn twice, as two components	One part counted as two, and never batched.
A component named 30" that measures 34"	The name was wrong and the geometry was right.
A 3D Warehouse scale figure left in the model	Appears in the part list as unidentified stock.

The acceptance test

The file is not finished when it looks right. It is finished when the script reads it cleanly — and you can check that yourself before sending anything.

```
SketchUp: Extensions > Developer > Ruby Console
load ".../ccp_extract.rb"
```

The file passes when

- ☐ The verdict line reads COMPONENTISED – ready to number
- ☐ No NO PROFILE MATCH rows, except genuinely non-metal parts such as a stone top
- ☐ No ASSUMED in the gauge column
- ☐ No ORPHAN definitions
- ☐ Every part measures the length it is meant to be cut
- ☐ The product appears as one top-level component

Duplicate part numbers are not a failure. When two components produce the same number it means they are the same part — which is the system working, and the finding you want. It means merge them in the model, not fix them.

The one-line version

DRAW EVERY PART SEPARATELY, SQUARE TO THE AXES, AT ITS TRUE CUT LENGTH. COPY RATHER THAN REDRAW. WRAP IT IN A PRODUCT COMPONENT. DO NOT NAME ANYTHING.